

Genetic evidence for multiple introduction events of raccoons (*Procyon lotor*) in Spain

Fernando Alda · María José Ruiz-López ·
Francisco José García · Matthew E. Gompper ·
Lori S. Eggert · Jesús T. García

Received: 12 December 2011 / Accepted: 21 August 2012 / Published online: 31 August 2012
© Springer Science+Business Media B.V. 2012

Abstract The common raccoon (*Procyon lotor*) is endemic to Central and North America, although non-native populations have become established around the world. In Spain, growing evidence of the introduction of raccoons has been reported across the country in the last decade, especially in Central Spain where the largest population is thought to occur. We used mitochondrial and microsatellite DNA data to investigate the genetics of invasive raccoons in Central Spain and to infer: the number of introduction events, the number of founders and the genetic variability of the introduced populations compared to a native population. We found that at least two

introduction events have occurred along the Jarama and Henares Rivers in Central Spain, which currently constitute two genetically differentiated subpopulations. In both localities the number of effective founders from a native population was estimated as 2–4 individuals. These newly founded populations have expanded and show evidence of incipient contact and reproduction between them. This may allow for an increase in the genetic variability and adaptive potential of the population(s), possibly increasing the difficulty of controlling this invasive species. Our results reveal the ability to longitudinally monitor the genetics of the raccoon range expansion and emphasize the urgent need to control the pet trade of potentially invasive species.

F. Alda · J. T. García
Instituto de Investigación en Recursos Cinegéticos
(CSIC-UCLM-JCCLM), Ronda de Toledo s/n,
13005 Ciudad Real, Spain

F. Alda (✉)
Smithsonian Tropical Research Institute, Apartado
0843-03092, Balboa, Ancón, Republic of Panama
e-mail: alda.fernando@gmail.com

M. J. Ruiz-López · L. S. Eggert
Division of Biological Sciences, University of Missouri,
Columbia, MO 65211, USA

F. J. García
BIOTA, C/Alcázar de San Juan, 28011 Madrid, Spain

M. E. Gompper
Department of Fisheries and Wildlife Sciences, University
of Missouri, Columbia, MO 65211, USA

Keywords Raccoon · *Procyon lotor* · Spain ·
Founder population size · Invasion · Genetic variation

Introduction

Most long-distance introductions of nonnative species to new areas are the direct or indirect result of human activities (Sakai et al. 2001). Historically, exotic vertebrate species were deliberately introduced for hunting, fishing, fur farming or biological control (Fenner and Fantini 1999). More recently, the introduction of species as an indirect consequence of the pet trade has become a concern (Westphal et al. 2008; Smith et al. 2009), with some pet releases or escapes

such as monk parakeets (*Myiopsitta monachus*) (Russello et al. 2008) and red-eared terrapins (*Trachemys scripta*) (Warwick 1991) resulting in well-known colonizations of cities. The incidence of these exotic species introductions has been positively related to the economic activity of each region, such as per capita income or population density (Dalmazzone 2002; Westphal et al. 2008), as well as to activities such as agriculture, logging, grazing and urbanization that enhance the establishment of exotics by creating disturbed sites for colonization (Sakai et al. 2001). While such correlations are not specific to species kept as pets, urban and highly populated areas are likely to hold a large number and diversity of pets which, if released, might become invasive species.

In Spain, the last decade has seen increased evidence of raccoons (*Procyon lotor*), a species native to North and Central America, primarily in or near large cities. The largest raccoon population is thought to occur in central Spain, around Madrid, although the species has already been reported in at least 28 localities across the country (García et al. 2012). Raccoons were first reported in 2003 from tracks, later confirmed by photo-trapping, within Parque Regional del Sureste (PRS; Fig. 1) (Barona and García-Román 2005). Since then raccoons have spread beyond the bounds of the park and are reproducing along more than 28 km of nearby riparian areas (see Materials and Methods). Thus, the species has proven capable of colonization and range expansion, reaching locally high densities and constituting a reproductive population <10 years after introduction (García 2007). This introduction thus presents a threat to the many protected native species upon which raccoons may predate (Bartoszewicz 2006; García et al. 2012), as well as an important health issue for animals and humans because of pathogens raccoons may host (Park et al. 2000; Hohmann et al. 2002).

How many animals have been introduced in Spain, how many times, and their origin is unknown, although evidence suggests that they derive from escapees or deliberate releases of pets (García et al. 2012). The latter is common for these animals, which become aggressive at sexual maturity (Kauhala 1996; Ikeda et al. 2004). The number of introductions and founders, as well as the genetic diversity of the source population, will influence the genetic diversity of the newly founded population, in turn affecting its ability to establish and spread, as high levels of additive

genetic variation are assumed to increase the potential of adaptation of an introduced species (Sakai et al. 2001; Lee 2002; Lavergne and Molofsky 2007; Dlugosch and Parker 2008). In general, invasive populations founded by a small number of individuals from a single source population show low genetic variation and small effective population sizes, whereas invasions derived from multiple introductions have high genetic variation and large effective population sizes. Indeed, when individuals from multiple source populations found a population the ensuing genetic variability can be higher than a typical native source population (Kolbe et al. 2004; Simberloff 2009; Funk et al. 2011). Such links between genetic variation, multiple introductions, and invasion success may influence management strategies for invasive species (Dlugosch and Parker 2008).

Genetic approaches provide the tools to gain information about the history of a population, and allow detection, tracking, and predictions about the future of populations (Alda et al. 2008; Dlugosch and Parker 2008; Kalinowski et al. 2010; Fitzpatrick et al. 2011). Thus, the main objective of this study was to understand the recent invasion history of the raccoon in Central Spain using a genetic approach that combines information from mitochondrial and nuclear DNA markers. Specifically, we assessed: (1) the minimum number of raccoon introduction events in central Spain, (2) the genetic variation and effective population size remaining after introduction compared to a North American native raccoon population, and (3) the relative effective founder size.

Materials and methods

Study area and sampling methods

The study area is located in the central Spanish Plateau within the well-developed drainage network of the Tajo river basin (Fig. 1). The majority of this area is composed of floodplain, characterized by marshy meadows, irrigated crops (*Zea mays*, *Triticum* spp.) and wooded riparian areas of willows (*Salix* spp.) and poplars (*Populus* spp.), which are used by an abundant community of aquatic birds, including some threatened species (García et al. 2012). Sampling was performed along the Jarama and Henares rivers which cross the Parque del Sureste (40°17'0"N, 3°30'0"W),

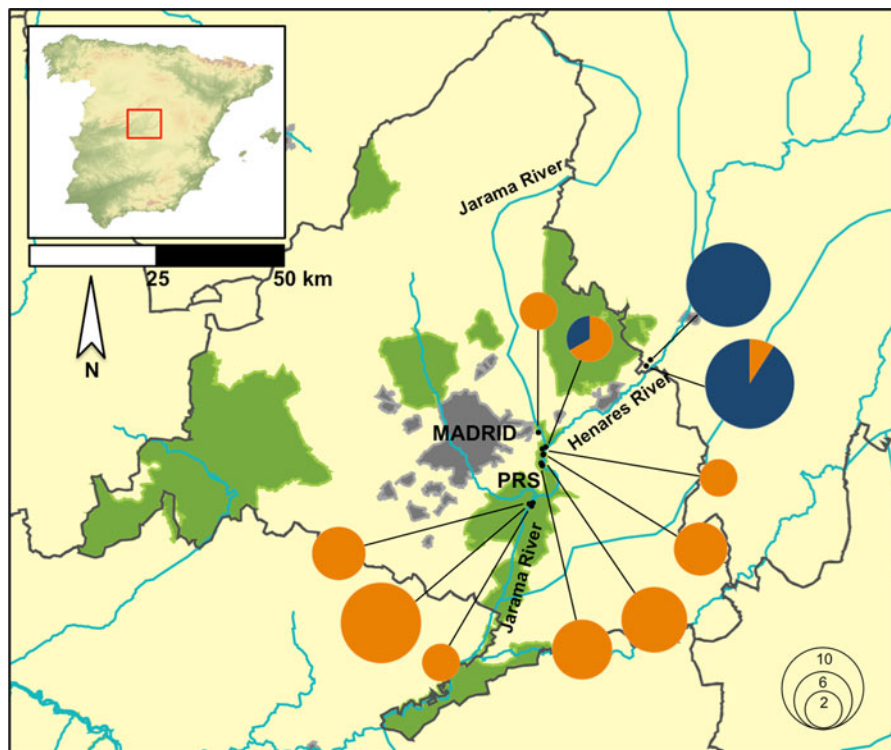


Fig. 1 Geographic distribution and frequency of haplotypes observed in raccoons sampled in Central Spain. The size of the pie charts is proportional to the number of individuals sampled per locality (from 2 to 11). Haplotype PLO2 is indicated in orange and PLO66 in blue. Grey lines are the borders of Spanish

provinces and light blue lines represent main rivers. Urban areas are indicated in grey and natural protected areas within the province of Madrid in green. PRS indicates Regional Natural Park Parque del Sureste. (Color figure online)

part of the European Union's Natura 2000 network of protected areas, in the southeastern part of the province of Madrid, ca. 20 km from the city of Madrid, up to the locality of Azuqueca de Henares in the province of Guadalajara, ca. 60 km northeast of Madrid (40°33'53"N, 3°16'05"W) (Fig. 1).

Raccoons were captured using home-made box traps baited with peanut butter. Trapping was conducted during intensive trapping sessions conducted between 2007 and 2010 as part of a broader effort to remove individuals, assess their parasite community and diet, and study the movements of animals using radio-telemetry (Aramburu et al. 2010; García et al. 2012). A sample of ear tissue was obtained from each animal captured following euthanasia by veterinarians from the regional administration (Spanish Wildlife Recovery Centres, CRAS). Tissue was preserved at -20°C until DNA extraction.

The origin of the raccoons introduced in Spain is unknown. Thus, to compare the genetic diversity and

relative effective population size of the introduced populations with that observed in a native population, we obtained data from a large raccoon population inhabiting the central United States, for which genetic diversity values are within the range of other studies in raccoon native populations (see Discussion and Cullingham et al. 2009; Dharmarajan et al. 2009; Côté et al. 2012). Raccoons were sampled between 2005 and 2007 near Columbia, Missouri, USA, from populations that have been the basis of a series of studies examining the molecular, population and disease ecology of raccoons in the region. We selected $n = 77$ unrelated individuals from 2 contiguous sites located ca. 20–30 km from the city of Columbia: the University of Missouri's Thomas S. Baskett Research and Education Center (38°46'20.659"N, 92°11'6.179"W) and United States Forest Service Mark Twain National Forest lands near County Road 354 (38°49'22.264"N, 92°8'14.331"W). Blood samples were collected from individuals via femoral venipuncture and used in

ensuing genetic analyses. Details of the Missouri raccoon population and on field methodologies are given elsewhere (Monello and Gompper 2007, 2009, 2010; Gompper et al. 2011; Monello and Gompper 2011; Wehtje and Gompper 2011).

DNA isolation and data collection

Genomic DNA was extracted from tissue or blood samples using a standard ammonium acetate precipitation protocol and DNeasy tissue kits (QIAGEN), respectively. A fragment of the mitochondrial control region of 467 bp was amplified in all Spanish raccoon samples using primers L15997 (5'-CAC CATTAGCACCCAAAGCT-3', Ward et al. 1991) and PLO_CRL1 (5'-CGCTTAAACTTATGTCCTG TAACC-3', Cullingham et al. 2008). PCR amplifications were carried out in 20 µl reactions containing: 1X PCR buffer (Biotools), 0.3 µM of each primer, 0.2 mM of each dNTP, 2 mM MgCl₂, 1U Taq polymerase (Biotools) and 1 µl DNA (ca. 25 ng/µl). Cycling profile consisted of an initial denaturation step of 5 min at 94 °C, 30 cycles of 30 s at 94 °C, 30 s at 60 °C, and 30 s at 72 °C, followed by a final extension step of 5 min at 72 °C. PCR-products were purified with Exonuclease I and Shrimp Alkaline Phosphatase enzymatic reactions (United States Biochemical). Purified reactions were sequenced in an ABI3130xl automated sequencer (Applied Biosystems) using the BigDye terminator v.3.1 kit (Applied Biosystems) with the same primers used for PCR.

All samples from Spain and the USA were genotyped for fourteen microsatellite loci: PLM03, PLM05, PLM06, PLM07, PLM08, PLM09, PLM10, PLM11, PLM12, PLM13, PLM14, PLM15, PLM16 and PLM17 (Siripunkaw et al. 2008). The microsatellites were co-amplified in two multiplex PCRs (Mix1: PLM05, PLM09, PLM10, PLM11, PLM15, PLM16; Mix2: PLM03, PLM06, PLM07, PLM08, PLM12, PLM13, PLM14, PLM17) following the QIAGEN Multiplex PCR kit protocol for 40 cycles and 58 °C of annealing temperature. Reactions were prepared in a final volume of 10 µl including: 5 µl of Qiagen 2X PCR Master Mix, 0.325 µl of 10X primer mix (2 µM each), 1 µl DNA (ca. 25 ng/µl), 0.8 mg/ml BSA and 2.875 µl of RNase-free H₂O. For the Spanish samples, fluorescently labeled PCR products were analyzed on an ABI3130xl DNA Analyzer (Applied Biosystems) and allele sizes were

determined using GeneMapper 3.7 software (Applied Biosystems). For the Missouri samples, fluorescently labeled PCR products were analyzed on an ABI 3730 DNA Analyzer (Applied Biosystems), and alleles were scored using GeneMarker 1.5 software (SoftGenetics, State College, PA, USA).

Genetic analysis

All control region sequences were checked and manually aligned using MacClade 4.08 (Maddison and Maddison 2000). DnaSP 5.0 (Librado and Rozas 2009) was used to determine the number of haplotypes (h) and variable sites (S), and to calculate haplotype (H_d) and nucleotide diversity (π).

Microsatellite genotype frequencies were tested for deviations from Hardy–Weinberg equilibrium using Genodive 2.0b20 (Meirmans and Van Tienderen 2004). Genetic diversity parameters: number of alleles (N_A), allelic richness (A_R), observed and expected heterozygosity (H_o and H_E) and inbreeding coefficient (F_{IS}) were estimated using FSTAT 2.9.3 (Goudet 1995).

Genetic differentiation among samples was first visualized with a Factorial Correspondence Analysis (FCA) using the program Genetix 4.02 (Belkhir et al. 2004). We then analyzed population genetic structure using the Bayesian model-based clustering method implemented in STRUCTURE 2.3.3 (Pritchard et al. 2000) under the admixture model and correlated allele frequencies (Falush et al. 2003). We performed 10 independent runs for each K value from $K = 1$ to $K = 6$. Each run had 5×10^5 iterations with a burn-in of 1×10^5 iterations. For each value of K , we averaged the log-likelihood values and calculated their associated posterior probabilities according to Bayes' Rule (Pritchard et al. 2009).

In addition, we estimated relatedness within and among the raccoon populations identified by STRUCTURE 2.3.3 in the previous analyses. The average pairwise relatedness coefficients (r_{xy} , Queller and Goodnight 1989) was calculated among all individuals in each of the raccoon populations utilizing the program GenAlEx 6.41 (Peakall and Smouse 2006). Given the low sample size, we calculated the average relatedness coefficient and 95 % confidence intervals (CI) using 1,000 permutations and 1,000 bootstraps. Means were reported $\pm$ standard deviation.

The size of the founding population was estimated by comparing the expected heterozygosity of the

Missouri native population with the expected heterozygosity in the Spanish population. If we assume that the introduced population in Spain was founded from N individuals from a native population like the one from Missouri, and the population in Spain grew rapidly after it was introduced, we can estimate the effective number of founding individuals N_f as

$$H_{\text{Spain}} = H_{\text{USA}} \left(1 - \frac{1}{2N_f} \right)$$

where H_{USA} is the heterozygosity in the native population of Missouri and H_{Spain} is the post-introduction heterozygosity in Spain (Eq. 6.3a, Hedrick 2005). A 95 % confidence interval for N_f was constructed by bootstrapping across loci 1,000 times.

The effects of a founder event on genetic variability were simulated using BottleSim 2.6 (Kuo and Janzen 2003), thereby estimating how many colonizers from a native population would be needed to explain the variability observed in the Spanish introduced population. We ran the model for 1,000 iterations with non-constant population size, random mating, generations overlapping by 80 %, reproductive maturity at 1 year of age, age of senescence at 15 years, and a sex ratio of 1:1. We constructed an initial population with the genetic characteristics of the Missouri population and subjected it to a bottleneck of 2–8 animals which grew exponentially for 7 years, corresponding to the time elapsed since the first detection of raccoons in Central Spain (Barona and García-Román 2005; García et al. 2012). Values of average N_A and H_E were plotted and compared with corresponding values observed in the Spanish populations. Statistically significant differences in the observed genetic diversity values among the simulated and the introduced populations were assessed by means of ANOVA and post hoc Tukey tests in STATISTICA 8.0 (StatSoft-Inc. 2007).

Also, current effective population sizes (N_e) were compared between native and introduced populations, since this is a crucial parameter in wildlife management because of its influence on population viability (Luikart et al. 2010). Two different one-point estimates methods (i.e. one sample per population, Hill 1981) were used to estimate N_e : the program ONeSAMP 1.2 (Tallmon et al. 2008), which uses an approximate Bayesian computation method, and the program LDNe (Waples and Do 2008), which estimates N_e based on linkage disequilibrium arising from genetic drift. A random mating system

was assumed, and all alleles with frequencies < 0.05 were excluded (Waples 2006).

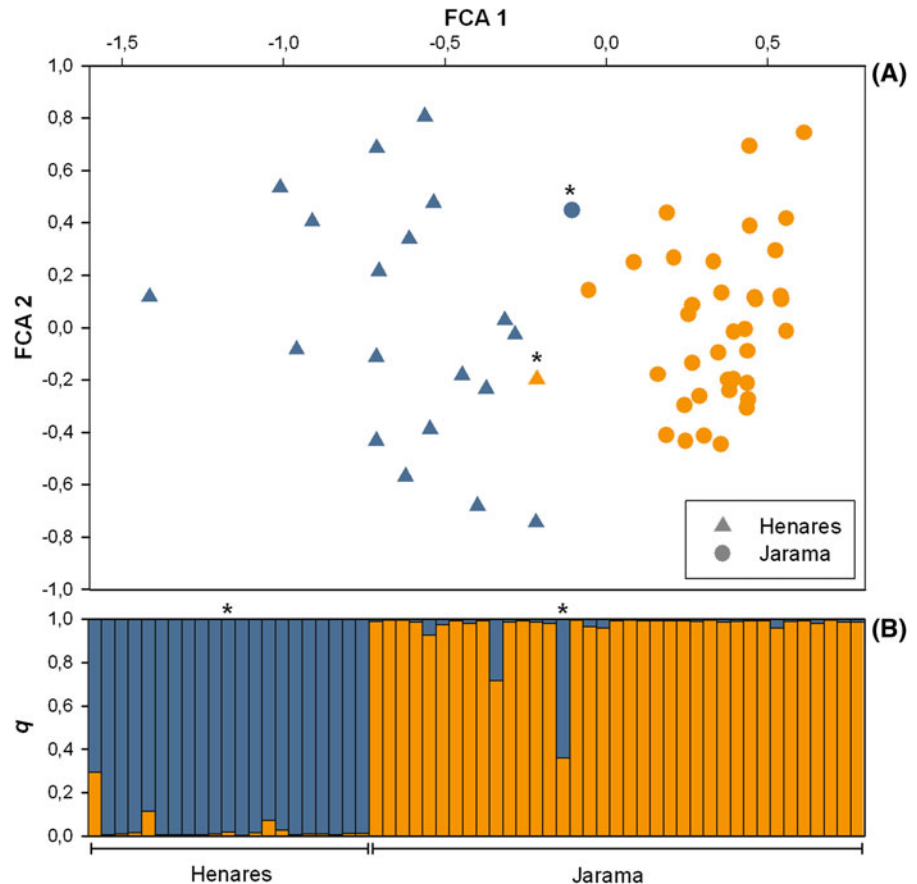
Results

A total of 58 introduced raccoons were captured in Spain. These include 37 individuals captured across 9 sites along the Jarama River and 21 individuals captured at two sites along the Henares River. Two mitochondrial haplotypes differing in one single nucleotide were found in the introduced Spanish population (PLO2 and PLO66, GeneBank Accession numbers: EF030393 and EF030409, respectively, Cullingham et al. 2008; Cullingham et al. unpublished), each of which was nearly fixed in the two main rivers that were sampled. Haplotype PLO2 was found in all individuals sampled along the Jarama River system, except for one individual captured at the conjunction of both rivers in the northern part of the PRS. In contrast, all individuals captured along the Henares River carried haplotype PLO66, with the exception of one individual (Fig. 1). Thus, both rivers showed low haplotype and nucleotide diversity ($H_d = 0.054 \pm 0.050$ and 0.095 ± 0.084 , and $\pi = 0.0001 \pm 0.0001$ and 0.0002 ± 0.0002 in the Jarama and Henares populations, respectively).

Analyses of nuclear microsatellite markers indicated that the Spanish raccoon population was clearly differentiated into two genetic groups. In the FCA two non-overlapping groups were observed corresponding to individuals sampled along the Jarama and Henares Rivers (Fig. 2a), which showed a high genetic differentiation ($F_{ST} = 0.150$, $P < 0.001$). Furthermore, the analysis performed in STRUCTURE revealed that the most likely number of genetic clusters was $K = 2$ (posterior probability $\cong 1$). Individuals within each river were assigned with high probability to one of the genetic clusters ($Q > 0.96$) and only one individual from the Jarama population was assigned with a higher probability ($q = 0.64$) to the genetic cluster from Henares population (Fig. 2b).

Genetic diversity at the nuclear level for all the Spanish samples was low (Table 1). Overall, the number of alleles per locus ranged from $N_A = 2$ to 6 (mean $N_A = 4.071 \pm 1.328$), whereas in the native population allele numbers ranged from $N_A = 6$ to 16 (mean $N_A = 9.929 \pm 2.673$) (Table 1). Allelic richness, permuted by the lowest number of individuals

Fig. 2 **a** Factorial component analysis performed for the raccoons sampled in Spain based on 14 microsatellite loci. Symbols represent individuals sampled in the Henares (triangles) and Jarama (circles) Rivers and colors indicate mitochondrial control region haplotypes PLO2 (orange) and PLO66 (blue). **b** Graphical summary of clustering analysis performed in STRUCTURE for the Spanish raccoons. Each individual is represented by a vertical line broken into two segments representing the estimated proportion of the individual assignment to each cluster. Blue corresponds to the Henares River cluster and orange to the Jarama River cluster. Asterisks denote those individuals sampled in the Henares and Jarama Rivers carrying mitochondrial haplotypes characteristic from the other river. (Color figure online)



genotyped in a population ($n = 21$ in Henares) and observed and expected heterozygosities were significantly lower in the two introduced populations than in the native population (Table 1, all ANOVA tests $F_{2,39} > 22.340$, $P < 0.001$, and post hoc Tukey Test: $P < 0.001$ for Jarama-Missouri and Henares-Missouri). Within the Spanish populations, genetic diversity was lower in Jarama, although the difference was non-significant (post hoc Tukey Test: $P > 0.235$ for all Jarama-Henares comparisons). Overall, all populations showed non-significant deviations from Hardy–Weinberg proportions (Table 1).

Average pairwise relatedness (r_{xy}) among individuals within each of the two introduced raccoon populations (Jarama $r_{xy} = 0.145 \pm 0.221$, 95 % CI: 0.162–0.127; Henares $r_{xy} = 0.152 \pm 0.229$, 95 % CI: 0.184–0.119) was higher than zero (i.e. than panmixia) and than the whole Spanish population average ($r_{xy} = -0.019 \pm 0.280$, 95 % CI: -0.005 to -0.032) suggesting that raccoons were not clustered randomly. Additionally, the two introduced population showed higher relatedness

than the native population from which the 77 unrelated individuals were taken, which was close to panmixia ($r_{xy} = -0.004 \pm 0.132$, 95 % CI: -0.003 to -0.006) (Ruiz-Lopez et al. unpublished).

Assuming that raccoons in Central Spain population expanded to a large size in the generation immediately after introduction, the estimated effective size of the founding population was $N_f = 1.944$ (95 % confidence interval 1.857–2.031) for Jarama and $N_f = 4.370$ (95 % confidence interval 4.027–4.712) for Henares. Similar values were obtained when considering other native raccoon populations as sources (Cullingham et al. 2009; Root et al. 2009; Côté et al. 2012), ranging from $N_f = 1.61$ –1.96 in Jarama to $N_f = 2.77$ –4.38 in Henares. Results of analyses using Bottlesim to estimate the minimum number of founders that could have been introduced in Spain, conditional to the average number of alleles per locus and the expected heterozygosity in the Spanish and Missouri populations, indicated that the average number of alleles and the expected heterozygosity

Table 1 Genetic diversity of invasive (Spain, consisting of Jarama and Henares Rivers) and native (USA, derived from Missouri) raccoon *Procyon lotor* populations based on 14 microsatellite loci

	PLM03	PLM05	PLM06	PLM07	PLM08	PLM09	PLM10	PLM11	PLM12	PLM13	PLM14	PLM15	PLM16	PLM17	Mean (SD)
Spain															
<i>n</i> = 58															
N_A	4	2	5	3	5	6	3	4	5	4	6	3	5	2	4.071 (1.328)
A_R	3.877	3.327	2	4.984	3	3.988	5.759	2.999	3.38	4.862	2.943	4.311	2	4.041	3.676 (1.089)
H_o	0.707	0.379	0.828	0.579	0.614	0.839	0.552	0.482	0.667	0.69	0.81	0.603	0.684	0.375	0.629 (0.149)
H_E	0.626	0.496	0.764	0.666	0.591	0.786	0.555	0.491	0.689	0.605	0.73	0.502	0.706	0.329	0.610 (0.127)
F_{IS}	-0.129	0.235	-0.083	0.131	-0.038	-0.068	0.005	0.019	0.033	-0.14	-0.109	-0.203	0.031	-0.14	-0.032 (0.118)
Jarama															
<i>n</i> = 37															
N_A	3	3	2	4	3	3	4	3	2	5	3	3	2	4	3.143 (0.864)
A_R	2.998	3	2	3.989	3	2.89	4	2.989	2	4.278	2.514	3	1.999	3.514	3.012 (0.740)
H_o	0.649	0.73	0.351	0.757	0.556	0.568	0.865	0.595	0.389	0.811	0.486	0.639	0.229	0.622	0.589 (0.180)
H_E	0.596	0.608	0.485	0.712	0.591	0.544	0.711	0.525	0.35	0.615	0.378	0.633	0.205	0.628	0.542 (0.144)
F_{IS}	-0.087	-0.199	0.276	-0.063	0.06	-0.043	-0.217	-0.133	-0.111	-0.318	-0.287	-0.01	-0.115	0.011	-0.087 (0.151)
Henares															
<i>n</i> = 21															
N_A	3	4	2	4	3	4	5	3	4	4	3	5	2	4	3.571 (0.938)
A_R	3	3.898	2	3.993	2.905	3.95	5	3	3.999	3.993	3	4.81	2	3.949	3.536 (0.914)
H_o	0.762	0.667	0.429	0.952	0.619	0.7	0.789	0.476	0.65	0.81	0.81	0.762	0.619	0.75	0.700 (0.138)
H_E	0.605	0.57	0.514	0.695	0.51	0.663	0.721	0.598	0.642	0.657	0.623	0.681	0.467	0.571	0.608 (0.075)
F_{IS}	-0.26	-0.169	0.167	-0.37	-0.215	-0.056	-0.095	0.203	-0.012	-0.232	-0.3	-0.119	-0.327	-0.313	-0.150 (0.177)
USA															
<i>n</i> = 77															
N_A	8	10	10	12	10	12	11	5	9	16	11	9	6	10	9.929 (2.674)
A_R	5.646	8.149	7.776	9.398	8.465	10.24	9.892	3.877	7.279	12.211	9.754	8.156	5.527	8.711	8.220 (2.160)
H_o	0.724	0.724	0.753	0.842	0.808	0.92	0.909	0.711	0.714	0.919	0.853	0.867	0.592	0.867	0.800 (0.099)
H_E	0.659	0.761	0.809	0.853	0.802	0.878	0.881	0.687	0.811	0.901	0.882	0.832	0.619	0.845	0.801 (0.089)
F_{IS}	-0.099	0.049	0.069	0.013	-0.008	-0.048	-0.032	-0.034	0.12	-0.02	0.032	-0.042	0.044	-0.026	0.002 (0.057)

Standard deviation is shown in parentheses. Bold values indicate significant departures from Hardy–Weinberg equilibrium ($P < 0.05$) n Number of samples, N_A number of alleles, A_R allelic richness standardized to the minimum sample size, H_o observed heterozygosity, H_E expected heterozygosity, F_{IS} inbreeding index

values were significantly different among populations with increasing number of founders (ANOVA $F_{6,91} = 64.272$, $P < 0.001$ and $F_{6,91} = 16.864$, $P < 0.001$ for N_A and H_E , respectively; post hoc Tukey Test: $P < 0.001$ for all comparisons). The average number of alleles per locus $N_A = 3.143 \pm 0.864$ observed for the Jarama population would have needed only 2 effective founders from a native raccoon population (post hoc Tukey Test: $P < 0.001$ for $N_f = 3$ -Jarama), while at least 3 effective founders would have been needed to render the average number of alleles per locus ($N_A = 3.571 \pm 0.938$) observed in the Henares population (post hoc Tukey Test: $P < 0.001$ for $N_f = 4$ -Henares) (Fig. 3). Considering the expected heterozygosity values, the simulations performed indicated that 2 effective founders were needed for the Jarama population (post hoc Tukey Test: $P = 0.042$ for $N_f = 3$ -Jarama) and 4 for the Henares population (post hoc Tukey Test: $P < 0.010$ for $N_f = 5$ -Henares) to explain their expected heterozygosity values ($H_E = 0.542 \pm 0.144$ and 0.608 ± 0.075 , respectively) (Fig. 3).

Current effective population size estimates were slightly different between methods, but congruent and with largely overlapping confidence intervals (Table 2). In all cases the invasive populations showed similar and much lower N_e values (mean values = 12.10–18.20) than the native population (mean values = 370.07–854.00), for which the LDNe method failed to detect any evidence of genetic drift between generations, thus rendering an infinite upper 95 % confidence interval.

Discussion

Our data supports the existence of two geographically and genetically differentiated populations of raccoons in central Spain that can be attributed to at least two independent introduction events (Funk et al. 2011). The two population nuclei are respectively distributed along two rivers to the east of the city of Madrid (the Jarama and Henares Rivers) and, at present, can be distinguished both by mitochondrial and nuclear allele frequencies.

Only two mitochondrial haplotypes were found in the introduced raccoons of Central Spain, which were nearly fixed in the two populations. PLO2 was found in 36 of 37 individuals (97 %) along the Jarama River and

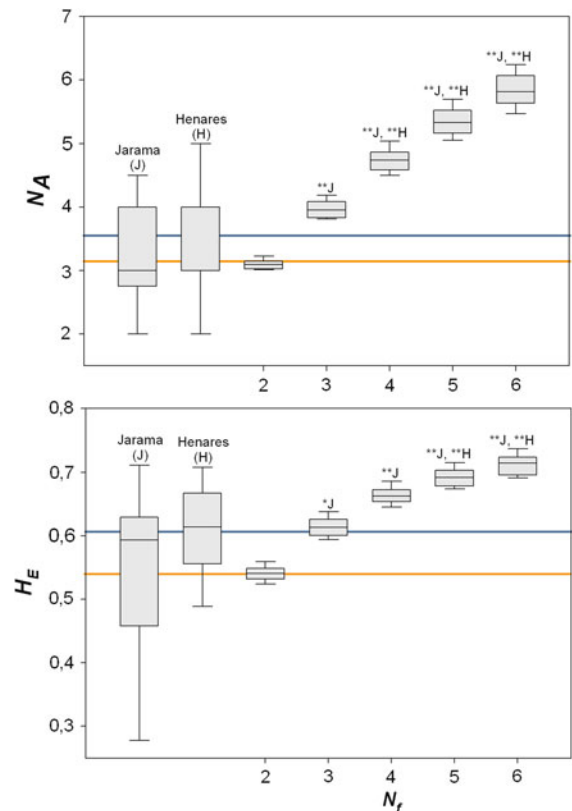


Fig. 3 Founder event simulations obtained using BottleSim. Effects of founder events of variable size, from 2 to 6 founders (N_f), on the average number of alleles per locus (N_A) and the expected heterozygosity (H_E) in newly founded populations growing for 7 years. Box plots represent median and 25th and 75th percentiles, and whiskers above and below the boxes indicate 90th and 10th percentiles of the observed values in raccoons from the Jarama (J) and Henares (H) Rivers and in each of the simulated scenarios. Horizontal lines represent observed mean values in the Jarama (orange) and Henares (blue) Rivers. Asterisks denote statistically significant differences between the observed values in each river and each simulation scenario (Post hoc Tukey Test: * $P < 0.05$; and ** $P < 0.01$). (Color figure online)

PLO66 was found in 20 of 21 individuals (95 %) sampled along the Henares River. As would be expected, these haplotypes represent a very small fraction of the genetic diversity observed in native populations. For instance, Cullingham et al. (2008) reported values of $H_d = 0.841$ – 0.969 and $\pi = 0.007$ – 0.136 , although at a much larger geographical scale. Similarly, the nuclear genetic diversity of the introduced population was significantly lower than in native populations, here represented by the Missouri population which showed diversity values consistent with studies from other regions in mainland North

Table 2 Effective population size (N_e) values estimated for the invasive (Spain, consisting of Jarama and Henares) and native (USA, consisting of Columbia MO) raccoon *P. lotor* populations

Method	Mean	Lower 95 % CI	Upper 95 % CI
<i>ONeSAMP</i>			
Jarama	15.38	13.21	19.69
Henares	18.16	14.92	23.54
USA	370.07	179.19	1,093.33
<i>LDNe</i>			
Jarama	12.10	7.80	55.40
Henares	18.20	14.50	55.40
USA	854.00	350.60	Infinite

America, albeit as assessed using different sets of marker (H_o from 0.79 to 0.85, Cullingham et al. 2009; Root et al. 2009; Côté et al. 2012).

In the native range, haplotype PLO2 is the most common haplotype in the wild and it is widespread across all the raccoon subspecies in North America (Cullingham et al. 2008). PLO66 is less frequent but is nonetheless widely distributed across the northeastern USA (Cullingham pers. comm.). Based on this evidence we cannot identify the origin of these raccoons in the wild, since both haplotypes are widespread and may even occur in sympatry. However, the striking difference in haplotype frequencies observed between rivers and the high level of genetic differentiation revealed by the microsatellite markers, without an evident geographic barrier, and compared to the low level of differentiation observed in the species native range ($F_{ST} = 0.0019\text{--}0.022$, Cullingham et al. 2008, 2009; Dharmarajan et al. 2009; Root et al. 2009; Côté et al. 2012), strongly supports the existence of at least two independent introduction events and likely two source populations (Funk et al. 2011). Thus, no less than two introduction events have occurred in the study region of Central Spain. Each founder event consisted of either a single independent maternal lineage, or multiple lineages that were subsequently lost through drift until fixation. Currently, only one individual in each river subpopulation carried the mitochondrial haplotype characteristic from the other. In the case of the Jarama population, this individual was found in the northern part of the river (Fig. 1) and was assigned with a 0.62 probability to the Henares genetic cluster (Fig. 2). Therefore, we could deduce that this individual is a first generation

descendant from a migrant of the Henares population. On the other hand, the individual found along the Henares with haplotype PLO2, characteristic of the Jarama population, showed no shared ancestry in its nuclear genome with the individuals from Jarama population (Fig. 2). This could mean that this individual comes from the Henares maternal lineage but has been subject to many generations of backcrossing in Jarama population. However, given the high genetic differentiation between the two populations it is unlikely that a sufficient number of generations have elapsed since the introduction of the raccoon in Spain to have erased all genetic signals from its nuclear genome ancestry. Thus, it is more likely that more than one mitochondrial haplotype was found in the founder population of Henares; one being the most common in the native range PLO2 and the other PLO66.

The existence of two introduction events currently conforming two genetically differentiated nuclei, both at nuclear and mitochondrial level, is consistent with our estimates of relatedness, which indicate that these nuclei might represent two closely related groups of individuals or family groups. This nonrandom spatial distribution of related individuals might be affected, among other factors, by the spatial distribution of resources (Dharmarajan et al. 2009; Wehtje and Gompper 2011). In urban and disturbed areas, raccoons can utilize small home ranges owing to the presence of highly aggregated anthropogenic resources (Prange et al. 2004; Dharmarajan et al. 2009; Wehtje and Gompper 2011) such as corn crops, which can be a major resource for the species (Rivest and Bergeron 1981) and which are abundant in the study region of Central Spain (García et al. 2012). This situation might have facilitated the use of small home ranges in the newly founded populations that together with a strong founder effect could explain both the high relatedness and the genetic structure observed at such low scale.

Both putative Spanish populations showed very small effective founder sizes. According to our simulations and frequency-based estimators, just two raccoons introduced from a wild population would be enough to represent the current genetic diversity in the Jarama population, as would 4 founder individuals in the Henares population. This pattern was also reflected in a slightly higher genetic variability in Henares than in Jarama, although the difference is statistically non-significant. However, this pattern is remarkable considering the much larger distribution of the Jarama

population, which occupies most of the PRS. Similarly, current N_e values were slightly higher in the Henares than in the Jarama population, but with highly overlapping confidence intervals. Overall, the invasive populations showed from 25 to 71-fold lower N_e values than those estimated for the native population.

Considering the limitations of our sample sizes and the nature of the invasive population, the precision of our estimates of the founding populations should be interpreted cautiously. However, taking into account that the presence of raccoon in Spain is very recent and that our study includes a large percentage ($\sim 75\%$) of the whole population in the area (García et al. 2012), we believe that our results offer a good representation of the population's true genetic diversity in the area. Therefore, these results are consistent with the potential for raccoons to become established and spread rapidly despite derivation from just a few founding individuals.

Further, it seems that the invasive potential of this taxon is not limited by the low diversity represented by the observed neutral genetic variability. Low genetic variability is not incompatible with successful invasion potential, because demographic bottlenecks may not eliminate all quantitative variation (Nei et al. 1975) and consequently many fitness-related traits could be retained as these are not lost as readily as individual alleles (Dlugosch and Parker 2008). Additionally, the characteristics of the region where the introduced population occurs in Spain, which is both highly urbanized and modified by intensive agricultural practices, are thought to boost rates of colonization and spread of invasive species (Sakai et al. 2001). The invasive capability of raccoons has already permitted the two subpopulations to come into contact, and at the intersection of the two rivers a possible first generation descendant of the two subpopulations was identified. This observation also supports the idea that rivers may act as corridors for this species in Spain (García et al. 2012).

The incipient contact between the two subpopulations raises a broader concern. Gene flow between introduced populations entails concerns for the control of invasive species as it can increase the genetic variability and associated adaptive potential of the individual populations (Dlugosch and Parker 2008). However, knowledge of the origin and genetic history of an introduced population can also serve as a landmark for future studies that can further evaluate

changes in the invasiveness of the species (Lizarralde et al. 2008). Such information would be crucial to create a monitoring plan for the population and for identifying management programs to effectively control this invasive species. If periodic introductions of new individuals deriving from the pet trade continue, control of raccoons in the field alone is likely to be of limited success, as even a low number of individuals will allow the population to rapidly recover. Thus it is essential to strictly control the pet trade for this species (García et al. 2012), perhaps by limiting breeding of captive animals and by genetically characterizing commercial stocks to facilitate identifying and tracking illegal or accidental releases and subsequent range expansions.

Acknowledgments The authors would like to thank M.J. Aramburu, Y. Cortés, L. García-Román, J.L. González, J. Herrera, M. Pérez, B. Pliego, B. Prieto and the staff of the Regional Park *Parque Regional del Sureste*, Comunidad Autónoma de Madrid, for their help during fieldwork. Capture authorizations were provided by the Comunidad Autónoma de Madrid according to EU laws. We also thank C.I. Cullingham for providing additional information on the geographical distribution of native raccoon haplotypes. Two anonymous referees provided useful suggestions that helped to improve the manuscript. This study was partially funded by the Regional Park *Parque Regional del Sureste* and the Comunidad Autónoma de Madrid. Sampling and genotyping of Missouri raccoons was supported by grants from the National Science Foundation (DEB-0347609 and DEB-0841654).

References

- Alda F, Inogés J, Alcaraz L, Oria J, Aranda A, Doadrio I (2008) Looking for the Iberian lynx in central Spain: a needle in a haystack? *Animal Conserv* 11:297–305
- Aramburu MJ, Cortés Y, García FJ, González JL, Herrera J, Pérez MJ, Pliego B, Prieto B (2010) Gestión de las poblaciones de mapache (*Procyon lotor*) en el Parque Regional del Sureste. Unpublished technical report. Comunidad de Madrid, Madrid
- Barona J, García-Román L (2005) Presencia de mapache (*Procyon lotor*) en el Parque Regional del Sureste: distribución actual y abundancia relativa [Presence of raccoons (*Procyon lotor*) in the Regional Park of Sureste (Madrid): current distribution and relative abundance]. In: Proceeding of the VIII national congress of the Spanish society for the conservation and study of mammals (SECEM), p 15, Huelva, Spain (in Spanish)
- Bartoszewicz M (2006) NOBANIS—invasive alien species fact sheet. *Procyon lotor*. Retrieved from www.nobanis.org on 31 March 2011
- Belkhir K, Borsa P, Chikhi L, Raufaste N, Bonhomme F (2004) GENETIX 4.05, logiciel sous Windows™ pour la

- g n tique des populations. Laboratoire G nome, Populations, Interactions, CNRS UMR 5000, Universit  de Montpellier II, Montpellier
- C t  H, Garant D, Robert K, Mainguy J, Pelletier F (2012) Genetic structure and rabies spread potential in raccoons: the role of landscape barriers and sex-biased dispersal. *Evol Appl* 5:393–404
- Cullingham CI, Kyle CJ, Pond BA, White BN (2008) Genetic structure of raccoons in eastern North America based on mtDNA: implications for subspecies designation and rabies disease dynamics. *Can J Zool* 86:947–958
- Cullingham CI, Kyle CJ, Pond BA, Rees EE, White BN (2009) Differential permeability of rivers to raccoon gene flow corresponds to rabies incidence in Ontario, Canada. *Mol Ecol* 18:43–53
- Dalmazzone S (2002) Economic factors affecting the vulnerability to biological invasions. In: Perrings C (ed) *The economics of biological invasions*. Edward Elgar, Cheltenham, UK, pp 17–30
- Dharmarajan G, Beasley JC, Fike JA, Rhodes OE (2009) Population genetic structure of raccoons (*Procyon lotor*) inhabiting a highly fragmented landscape. *Can J Zool* 87:814–824
- Dlugosch KM, Parker IM (2008) Founding events in species invasions: genetic variation, adaptive evolution, and the role of multiple introductions. *Mol Ecol* 17:431–449
- Falush D, Stephens M, Pritchard JK (2003) Inference of population structure using multilocus genotype data: linked loci and correlated allele frequencies. *Genetics* 164:1567–1587
- Fenner F, Fantini B (1999) Biological control of vertebrate pests: the history of myxomatosis; an experiment in evolution. CABI Publishing, Wallingford
- Fitzpatrick BM, Fordyce JA, Niemiller ML, Reynolds RG (2011) What can DNA tell us about biological invasions? *Biol Invasions* 14:245–253
- Funk WC, Garcia TS, Cortina GA, Hill RH (2011) Population genetics of introduced bullfrogs, *Rana (Lithobates) catesbeianus*, in the Willamette Valley, Oregon, USA. *Biol Invasions* 13:651–658
- Garc a FJ (2007) Gesti n de las poblaciones de mapaches (*Procyon lotor*) en la Comunidad de Madrid [Management of Raccoon populations (*Procyon lotor*) in the Comunidad Aut noma de Madrid]. In: *Proceedings of the VIII National Congress of the Spanish Society for the Conservation and Study of Mammals (SECEM)*, p 79, Huelva, Spain (in Spanish)
- Garc a JT, Garc a FJ, Alda F, Gonz lez JL, Aramburu MJ, Cort s Y, Prieto B, Pliego B, P rez M, Herrera J, Garc a-Rom n L (2012) Recent invasion and reproduction of the Raccoon (*Procyon lotor*) in Spain. *Biol Invasions* 14:1305–1310
- Gompper ME, Monello RJ, Eggert LS (2011) Genetic variability and viral seroconversion in an outcrossing vertebrate population. *Proc R Soc Lond B* 278:204–210
- Goudet J (1995) FSTAT (version 1.2): a computer program to calculate F-statistics. *J Hered* 86:485–486
- Hedrick PW (2005) *Genetics of populations*. Jones and Bartlett, Boston, MA
- Hill WG (1981) Estimation of effective population size from data on linkage disequilibrium. *Genet Res* 38:209–216
- Hohmann U, Voight S, Andreas U (2002) Raccoons take the offensive. A current assessment. In: Kowarik I, Starfinger U (eds) *Biologische invasionen, herausforderung zum Handeln? Neobiota*, pp 191–192
- Ikeda T, Asano M, Matoba Y, Abe G (2004) Present status of invasive alien raccoon and its impact in Japan. *Glob Environ Res* 8:125–131
- Kalinowski ST, Muhlfeld CC, Guy CS, Cox B (2010) Founding population size of an aquatic invasive species. *Conserv Genet* 11:2049–2053
- Kauhala K (1996) Introduced carnivores in Europe with special reference to central and northern Europe. *Wildl Biol* 2:197–204
- Kolbe JJ, Glor RE, Schettino LR, Lara AC, Larson A, Losos JB (2004) Genetic variation increases during biological invasion by a Cuban Lizard. *Nature* 431:177–181
- Kuo CH, Janzen FJ (2003) BottleSim: a bottleneck simulation program for long-lived species with overlapping generations. *Mol Ecol Notes* 3:669–673
- Lavergne S, Molofsky J (2007) Increased genetic variation and evolutionary potential drive the success of an invasive grass. *Proc Natl Acad Sci USA* 104:3883–3888
- Lee CE (2002) Evolutionary genetics of invasive species. *Trends Ecol Evol* 17:386–391
- Librado P, Rozas J (2009) DnaSP v5: a software for comprehensive analysis of DNA polymorphism data. *Bioinformatics* 25:1451–1452
- Lizarralde MS, Bailliet G, Poljak S, Fasanella M, Giulivi C (2008) Assessing genetic variation and population structure of invasive North American beaver (*Castor canadensis* Kuhl, 1820) in Tierra del Fuego (Argentina). *Biol Invasions* 10:673–683
- Luikart G, Ryman N, Tallmon DA, Schwartz MK, Allendorf FW (2010) Estimation of census and effective population sizes: the increasing usefulness of DNA-based approaches. *Conserv Genet* 11:355–373
- Maddison DR, Maddison WP (2000) *MacClade 4: Analysis of phylogeny and character evolution*. Sinauer Associates, Sunderland, MA
- Meirmans PG, Van Tienderen PH (2004) GENOTYPE and GENODIVE: two programs for the analysis of genetic diversity of asexual organisms. *Mol Ecol Notes* 4:792–794
- Monello RJ, Gompper ME (2007) Biotic and abiotic predictors of tick (*Dermacentor variabilis*) abundance and engorgement on free-ranging raccoons (*Procyon lotor*). *Parasitology* 134:2053–2062
- Monello RJ, Gompper ME (2009) Relative importance of demographics, locale, and seasonality underlying louse and flea parasitism of raccoons (*Procyon lotor*). *J Parasitol* 95:56–62
- Monello RJ, Gompper ME (2010) Differential effects of experimental increases in sociality on ectoparasites of free-ranging raccoons. *J Anim Ecol* 79:602–609
- Monello RJ, Gompper ME (2011) Effects of resource availability and social aggregation on the species richness of raccoon endoparasite infracommunities. *Oikos* 120:1427–1433
- Nei M, Maruyama T, Chakraborty R (1975) The bottleneck effect and genetic variability in populations. *Evolution* 29:1–10

- Park SY, Glaser C, Murray WJ, Kazacos KR, Rowley HA, Fredrick DR, Bass N (2000) Raccoon roundworm (*Baylisascaris procyonis*) encephalitis: case report and field investigation. *Pediatrics* 106:E556
- Peakall R, Smouse PE (2006) GENALEX 6: genetic analysis in excel. Population genetic software for teaching and research. *Mol Ecol Notes* 6:288–295
- Prange S, Gehrt SD, Wiggers EP (2004) Influences of anthropogenic resources on raccoon (*Procyon lotor*) movements and spatial distribution. *J Mammal* 85:483–490
- Pritchard JK, Stephens M, Donnelly P (2000) Inference of population structure using multilocus genotype data. *Genetics* 155:945–959
- Pritchard JK, Wen X, Falush D (2009) Documentation for *structure* software: version 2.3. Department of Human Genetics, University of Chicago
- Queller DC, Goodnight KF (1989) Estimating relatedness using genetic markers. *Evolution* 43:258–275
- Rivest P, Bergeron J-M (1981) Density, food habits, and economic importance of raccoons (*Procyon lotor*) in Quebec agrosystems. *Can J Zool* 59:1755–1762
- Root JJ, Puskas RB, Fischer JW, Swope CB, Neubaum MA, Reeder SA, Piaggio AJ (2009) Landscape genetics of raccoons (*Procyon lotor*) associated with ridges and valleys of Pennsylvania: implications for oral rabies vaccination programs. *Vector-Borne Zoonot* 9:583–588
- Russello MA, Avery ML, Wright TF (2008) Genetic evidence link invasive monk parakeet populations in the United States to the international pet trade. *BMC Evol Biol* 8:217
- Sakai AK, Allendorf FW, Holt JS, Lodge DM, Molofsky J, With KA, Baughman S, Cabin RJ, Cohen JE, Ellstrand NC, McCauley DE, O'Neil P, Parker IM, Thompson JN, Weller SG (2001) The population biology of invasive species. *Annu Rev Ecol Syst* 32:305–332
- Simberloff D (2009) The role of propagule pressure in biological invasions. *Annu Rev Ecol Evol Syst* 40:81–102
- Siripunkaw C, Kongrit C, Faries KM, Monello RJ, Gompper ME, Eggert LS (2008) Isolation and characterization of polymorphic microsatellite loci in the raccoon (*Procyon lotor*). *Mol Ecol Resour* 8:199–201
- Smith KF, Behrens M, Schloegel LM, Marano N, Burgiel S, Daszak P (2009) Reducing the risks of the wildlife trade. *Science* 324:594–595
- StatSoft-Inc (2007) STATISTICA (data analysis software system) version 8. www.statsoft.com
- Tallmon DA, Koyuk A, Luikart G, Beaumont MA (2008) ONE-SAMP: a program to estimate effective population size using approximate Bayesian computation. *Mol Ecol Resour* 8:299–301
- Waples RS (2006) A bias correction for estimates of effective population size based on linkage disequilibrium at unlinked gene loci. *Conserv Genet* 7:167–184
- Waples RS, Do C (2008) LDNe: a program for estimating effective population size from data on linkage disequilibrium. *Mol Ecol Resour* 8:753–756
- Ward RH, Frazier BL, Dew-Jager K, Pääbo S (1991) Extensive mitochondrial diversity within a single Amerindian tribe. *Proc Natl Acad Sci USA* 88:8720–8724
- Warwick C (1991) Conservation of red-eared terrapins *Trachemys scripta elegans*: threats from international pet and culinary markets. *BCG Testudo* 3:34–44
- Wehtje M, Gompper ME (2011) Effects of an experimentally clumped resource on raccoon (*Procyon lotor*) home range use. *Wildl Biol* 17:25–32
- Westphal MI, Browne M, MacKinnon K, Noble I (2008) The link between international trade and the global distribution of invasive alien species. *Biol Invasions* 10:391–398